IoT-Enabled Smart Bus Route Monitoring and Revenue Management System

R26-ISE-004





INTRODUCTION



A smart digital transport management system designed to improve the reliability and efficiency of Sri Lanka's public bus services. The system uses IoT-based GPS devices installed in buses to capture real-time location, trip events, and route compliance data. A cloud platform processes this information to monitor bus operations and detect route deviations. It also provides dashboards for transport authorities and mobile apps for bus operators. By combining IoT tracking, automated scheduling, and policy monitoring, the system improves service reliability and operational transparency.

RESEARCH PROBLEM?

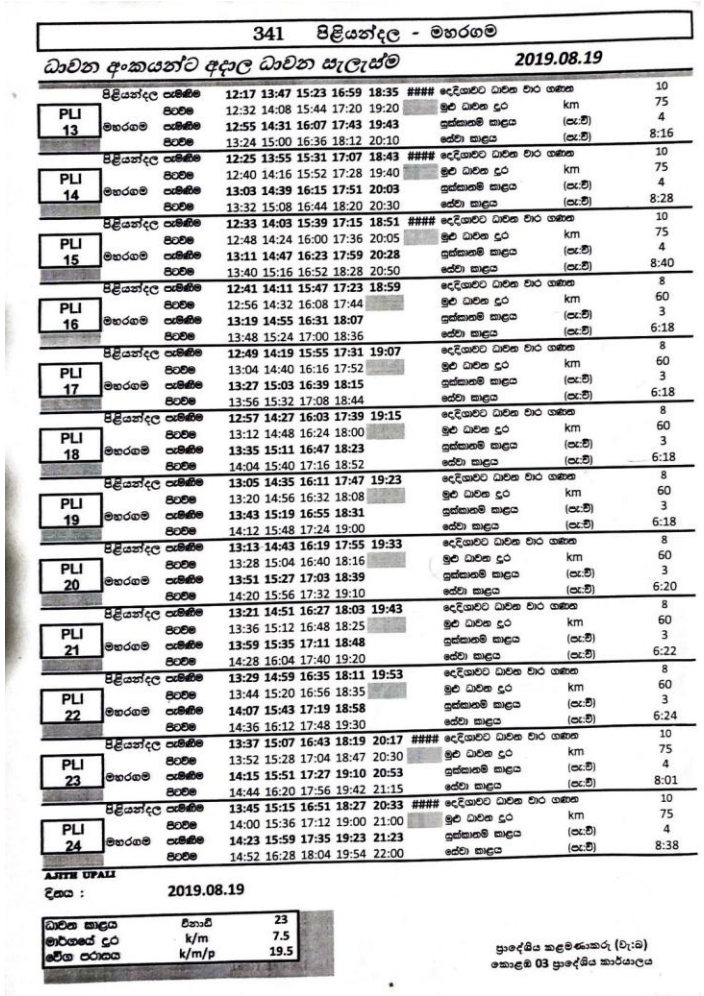
- Lack of public transportation availability during off peak hours
- Government's fixed permit payment system is losing potential income
- Buses tend to deviate from their permitted route

OBJECTIVES

- Improve the availability and reliability of public transportation during off peak times.
- Increase permit fee revenue
- Eliminating deviations from permitted route
- Update bus owners and drivers of their service incomes and expenses



BUS Schedules



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ධාවන අංකයන්ට අදාල ධාවන සැලැස්ම 2019.08.19

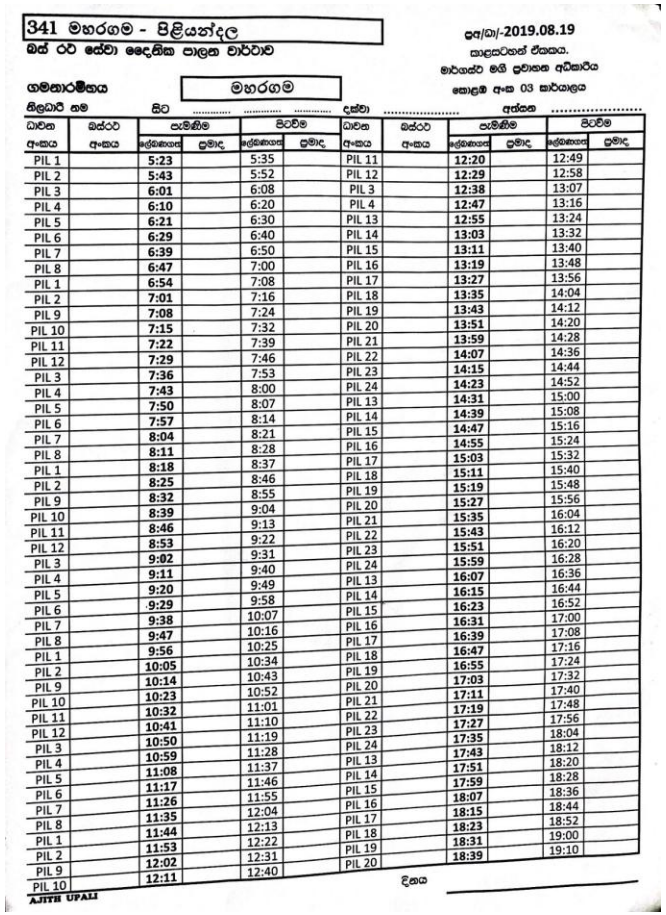
PLI 13	පිළියන්දල	පැමිණීම	12:17	13:47	15:23	16:59	18:35	####	දෙදිගාවට ධාවන වාර ගණන		10
		පිටවීම	12:32	14:08	15:44	17:20	19:20		මුළු ධාවන දුර	km	75
	මහරගම	පැමිණීම	12:55	14:31	16:07	17:43	19:43		ක්‍රක්‍රාන්ත කාලය	(පැ:ම)	4
		පිටවීම	13:24	15:00	16:36	18:12	20:10		සේවා කාලය	(පැ:ම)	8:16
PLI 14	පිළියන්දල	පැමිණීම	12:25	13:55	15:31	17:07	18:43	####	දෙදිගාවට ධාවන වාර ගණන		10
		පිටවීම	12:40	14:16	15:52	17:28	19:40		මුළු ධාවන දුර	km	75
	මහරගම	පැමිණීම	13:03	14:39	16:15	17:51	20:03		ක්‍රක්‍රාන්ත කාලය	(පැ:ම)	4
		පිටවීම	13:32	15:08	16:44	18:20	20:30		සේවා කාලය	(පැ:ම)	8:28
PLI 15	පිළියන්දල	පැමිණීම	12:33	14:03	15:39	17:15	18:51	####	දෙදිගාවට ධාවන වාර ගණන		10
		පිටවීම	12:48	14:24	16:00	17:36	20:05		මුළු ධාවන දුර	km	75
	මහරගම	පැමිණීම	13:11	14:47	16:23	17:59	20:28		ක්‍රක්‍රාන්ත කාලය	(පැ:ම)	4
		පිටවීම	13:40	15:16	16:52	18:28	20:50		සේවා කාලය	(පැ:ම)	8:40
PLI 16	පිළියන්දල	පැමිණීම	12:41	14:11	15:47	17:23	18:59		දෙදිගාවට ධාවන වාර ගණන		8
		පිටවීම	12:56	14:32	16:08	17:44			මුළු ධාවන දුර	km	60
	මහරගම	පැමිණීම	13:19	14:55	16:31	18:07			ක්‍රක්‍රාන්ත කාලය	(පැ:ම)	3
		පිටවීම	13:48	15:24	17:00	18:36			සේවා කාලය	(පැ:ම)	6:18
PLI 17	පිළියන්දල	පැමිණීම	12:49	14:19	15:55	17:31	19:07		දෙදිගාවට ධාවන වාර ගණන		8
		පිටවීම	13:04	14:40	16:16	17:52			මුළු ධාවන දුර	km	60
	මහරගම	පැමිණීම	13:27	15:03	16:39	18:15			ක්‍රක්‍රාන්ත කාලය	(පැ:ම)	3
		පිටවීම	13:56	15:32	17:08	18:44			සේවා කාලය	(පැ:ම)	6:18
PLI 18	පිළියන්දල	පැමිණීම	12:57	14:27	16:03	17:39	19:15		දෙදිගාවට ධාවන වාර ගණන		8
		පිටවීම	13:12	14:48	16:24	18:00			මුළු ධාවන දුර	km	60
	මහරගම	පැමිණීම	13:35	15:11	16:47	18:23			ක්‍රක්‍රාන්ත කාලය	(පැ:ම)	3
		පිටවීම	14:04	15:40	17:16	18:52			සේවා කාලය	(පැ:ම)	6:18
PLI 19	පිළියන්දල	පැමිණීම	13:05	14:35	16:11	17:47	19:23		දෙදිගාවට ධාවන වාර ගණන		8
		පිටවීම	13:20	14:56	16:32	18:08			මුළු ධාවන දුර	km	60
	මහරගම	පැමිණීම	13:43	15:19	16:55	18:31			ක්‍රක්‍රාන්ත කාලය	(පැ:ම)	3
		පිටවීම	14:12	15:48	17:24	19:00			සේවා කාලය	(පැ:ම)	6:18
PLI 20	පිළියන්දල	පැමිණීම	13:13	14:43	16:19	17:55	19:33		දෙදිගාවට ධාවන වාර ගණන		8
		පිටවීම	13:28	15:04	16:40	18:16			මුළු ධාවන දුර	km	60
	මහරගම	පැමිණීම	13:51	15:27	17:03	18:39			ක්‍රක්‍රාන්ත කාලය	(පැ:ම)	3
		පිටවීම	14:20	15:56	17:32	19:10			සේවා කාලය	(පැ:ම)	6:20
PLI 21	පිළියන්දල	පැමිණීම	13:21	14:51	16:27	18:03	19:43		දෙදිගාවට ධාවන වාර ගණන		8
		පිටවීම	13:36	15:12	16:48	18:25			මුළු ධාවන දුර	km	60
	මහරගම	පැමිණීම	13:59	15:35	17:11	18:48			ක්‍රක්‍රාන්ත කාලය	(පැ:ම)	3
		පිටවීම	14:28	16:04	17:40	19:20			සේවා කාලය	(පැ:ම)	6:22
PLI 22	පිළියන්දල	පැමිණීම	13:29	14:59	16:35	18:11	19:53		දෙදිගාවට ධාවන වාර ගණන		8
		පිටවීම	13:44	15:20	16:56	18:35			මුළු ධාවන දුර	km	60
	මහරගම	පැමිණීම	14:07	15:43	17:19	18:58			ක්‍රක්‍රාන්ත කාලය	(පැ:ම)	3
		පිටවීම	14:36	16:12	17:48	19:30			සේවා කාලය	(පැ:ම)	6:24
PLI 23	පිළියන්දල	පැමිණීම	13:37	15:07	16:43	18:19	20:17	####	දෙදිගාවට ධාවන වාර ගණන		10
		පිටවීම	13:52	15:28	17:04	18:47	20:30		මුළු ධාවන දුර	km	75
	මහරගම	පැමිණීම	14:15	15:51	17:27	19:10	20:53		ක්‍රක්‍රාන්ත කාලය	(පැ:ම)	4
		පිටවීම	14:44	16:20	17:56	19:42	21:15		සේවා කාලය	(පැ:ම)	8:01
PLI 24	පිළියන්දල	පැමිණීම	13:45	15:15	16:51	18:27	20:33	####	දෙදිගාවට ධාවන වාර ගණන		10
		පිටවීම	14:00	15:36	17:12	19:00	21:00		මුළු ධාවන දුර	km	75
	මහරගම	පැමිණීම	14:23	15:59	17:35	19:23	21:23		ක්‍රක්‍රාන්ත කාලය	(පැ:ම)	4
		පිටවීම	14:52	16:28	18:04	19:54	22:00		සේවා කාලය	(පැ:ම)	8:38

AJITH UPALI

දිනය : 2019.08.19

ධාවන කාලය	පිනාඩි	23
ආවරණය වූ දුර	k/m	7.5
මධ්‍යම පරාසය	k/m/p	19.5

ප්‍රාදේශිය කළමණාකරු (වැ:බ)
කොළඹ 03 ප්‍රාදේශිය කාර්යාලය



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ඔස් රථ සේවා දෛනික පාලන වාර්තාව

ප්‍රෙ/ධා/-2019.08.19
කාලසටහන් ඒකකය.
මාර්ගස්ථ මගී ප්‍රවාහන අධිකාරිය
කොළඹ අංක 03 කාර්යාලය

ගමනාරම්භය මහරගම

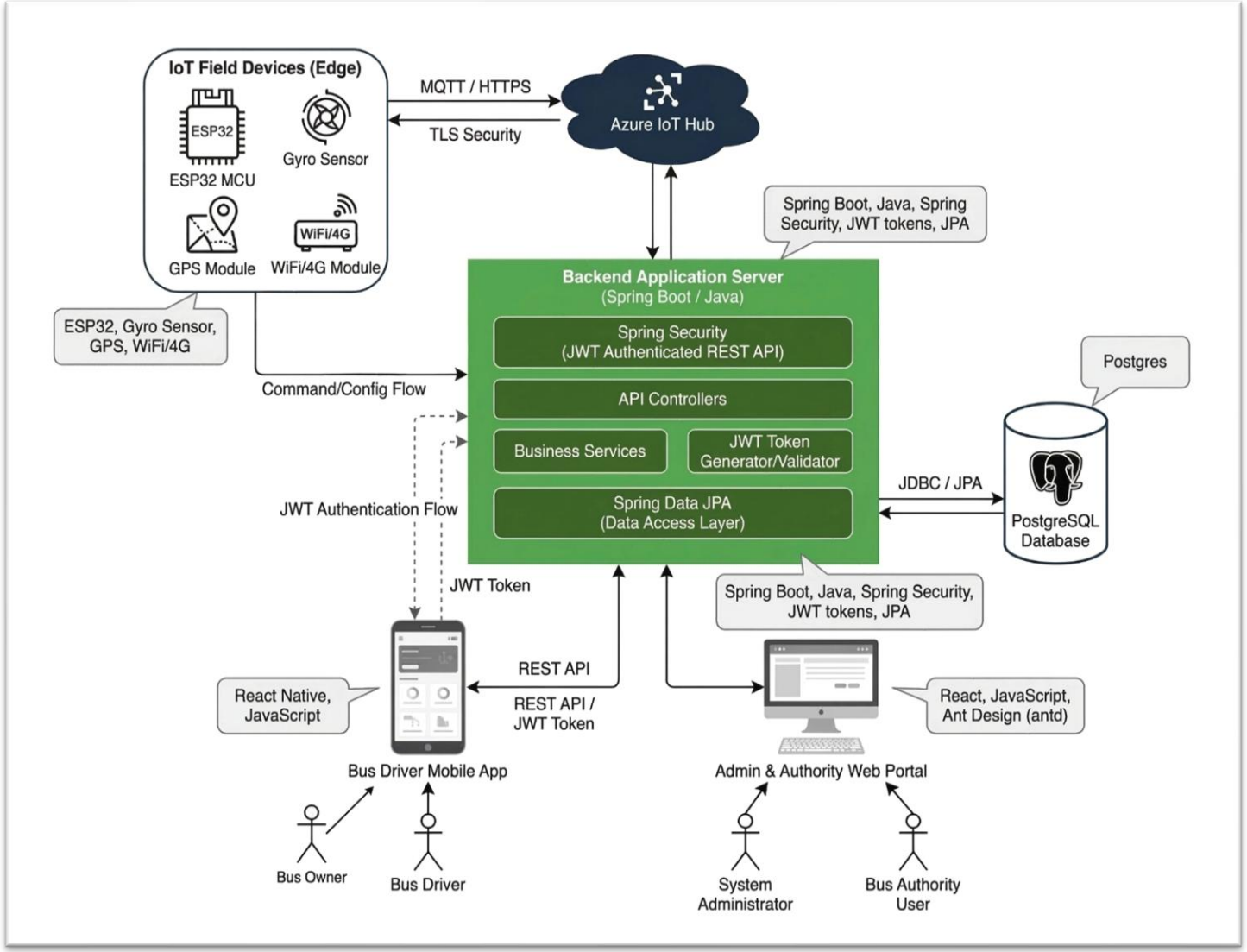
නිලධාරී නම සිට ..... දක්වා ..... අත්සන .....

ධාවන අංකය	බස්රථ අංකය	පැමිණීම		පිටවීම		ධාවන අංකය	බස්රථ අංකය	පැමිණීම		පිටවීම	
		ඇල්ලෙනගහ	ප්‍රමාද	ඇල්ලෙනගහ	ප්‍රමාද			ඇල්ලෙනගහ	ප්‍රමාද	ඇල්ලෙනගහ	ප්‍රමාද
PIL 1		5:23		5:35		PIL 11		12:20		12:49	
PIL 2		5:43		5:52		PIL 12		12:29		12:58	
PIL 3		6:01		6:08		PIL 3		12:38		13:07	
PIL 4		6:10		6:20		PIL 4		12:47		13:16	
PIL 5		6:21		6:30		PIL 13		12:55		13:24	
PIL 6		6:29		6:40		PIL 14		13:03		13:32	
PIL 7		6:39		6:50		PIL 15		13:11		13:40	
PIL 8		6:47		7:00		PIL 16		13:19		13:48	
PIL 1		6:54		7:08		PIL 17		13:27		13:56	
PIL 2		7:01		7:16		PIL 18		13:35		14:04	
PIL 9		7:08		7:24		PIL 19		13:43		14:12	
PIL 10		7:15		7:32		PIL 20		13:51		14:20	
PIL 11		7:22		7:39		PIL 21		13:59		14:28	
PIL 12		7:29		7:46		PIL 22		14:07		14:36	
PIL 3		7:36		7:53		PIL 23		14:15		14:44	
PIL 4		7:43		8:00		PIL 24		14:23		14:52	
PIL 5		7:50		8:07		PIL 13		14:31		15:00	
PIL 6		7:57		8:14		PIL 14		14:39		15:08	
PIL 7		8:04		8:21		PIL 15		14:47		15:16	
PIL 8		8:11		8:28		PIL 16		14:55		15:24	
PIL 1		8:18		8:37		PIL 17		15:03		15:32	
PIL 2		8:25		8:46		PIL 18		15:11		15:40	
PIL 9		8:32		8:55		PIL 19		15:19		15:48	
PIL 10		8:39		9:04		PIL 20		15:27		15:56	
PIL 11		8:46		9:13		PIL 21		15:35		16:04	
PIL 12		8:53		9:22		PIL 22		15:43		16:12	
PIL 3		9:02		9:31		PIL 23		15:51		16:20	
PIL 4		9:11		9:40		PIL 24		15:59		16:28	
PIL 5		9:20		9:49		PIL 13		16:07		16:36	
PIL 6		9:29		9:58		PIL 14		16:15		16:44	
PIL 7		9:38		10:07		PIL 15		16:23		16:52	
PIL 8		9:47		10:16		PIL 16		16:31		17:00	
PIL 1		9:56		10:25		PIL 17		16:39		17:08	
PIL 2		10:05		10:34		PIL 18		16:47		17:16	
PIL 9		10:14		10:43		PIL 19		16:55		17:24	
PIL 10		10:23		10:52		PIL 20		17:03		17:32	
PIL 11		10:32		11:01		PIL 21		17:11		17:40	
PIL 12		10:41		11:10		PIL 22		17:19		17:48	
PIL 3		10:50		11:19		PIL 23		17:27		17:56	
PIL 4		10:59		11:28		PIL 24		17:35		18:04	
PIL 5		11:08		11:37		PIL 13		17:43		18:12	
PIL 6		11:17		11:46		PIL 14		17:51		18:20	
PIL 7		11:26		11:55		PIL 15		17:59		18:28	
PIL 8		11:35		12:04		PIL 16		18:07		18:36	
PIL 1		11:44		12:13		PIL 17		18:15		18:44	
PIL 2		11:53		12:22		PIL 18		18:23		18:52	
PIL 9		12:02		12:31		PIL 19		18:31		19:00	
PIL 10		12:11		12:40		PIL 20		18:39		19:10	

AJITH UPALI

දිනය

SYSTEM DIAGRAM





IT22228208|W S H R GUNASEKARA

Distance-based Revenue Management Component

Distance-based Revenue Management Component



Research Problem:

The government doesn't have enough resources increase the availability and reliability of public transportation to ensure 24-hour service.



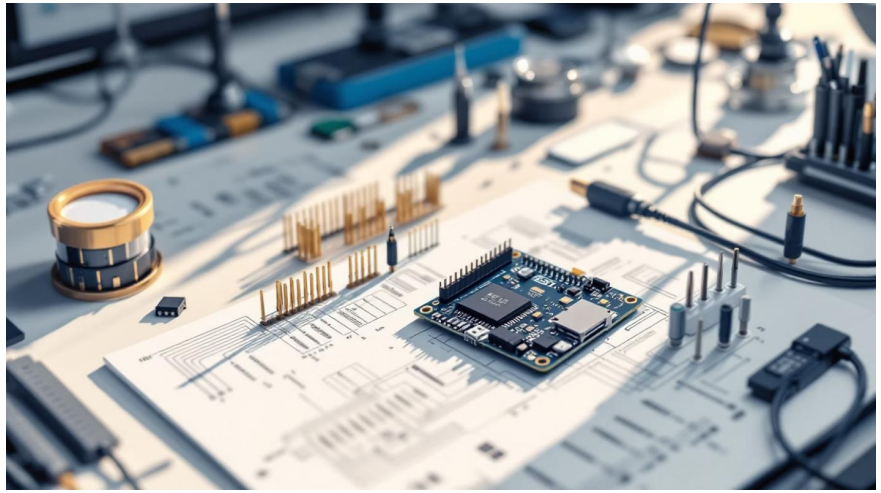
Specific Objective:

Increase government's revenue via a route permit fee calculation system that is fair for both transportation authorities and private bus owners.

Sub Objectives

- Develop an innovative device that tracks vehicle's location, speed and aggressiveness.
- Develop a distance-based revenue incentive model.
- Implement a commission payment function to encourage off-peak time service. .

Distance-based Revenue Management Component



Technologies

Hardware	Software
• ESP32 Microcontroller • NEO-6M GPS module • MPU6050 IMU Sensor • 5V Voltage Regulator	• Arduino IDE • Azure IoT Hub • Spring Boot • PostgreSQL • React

Methodology

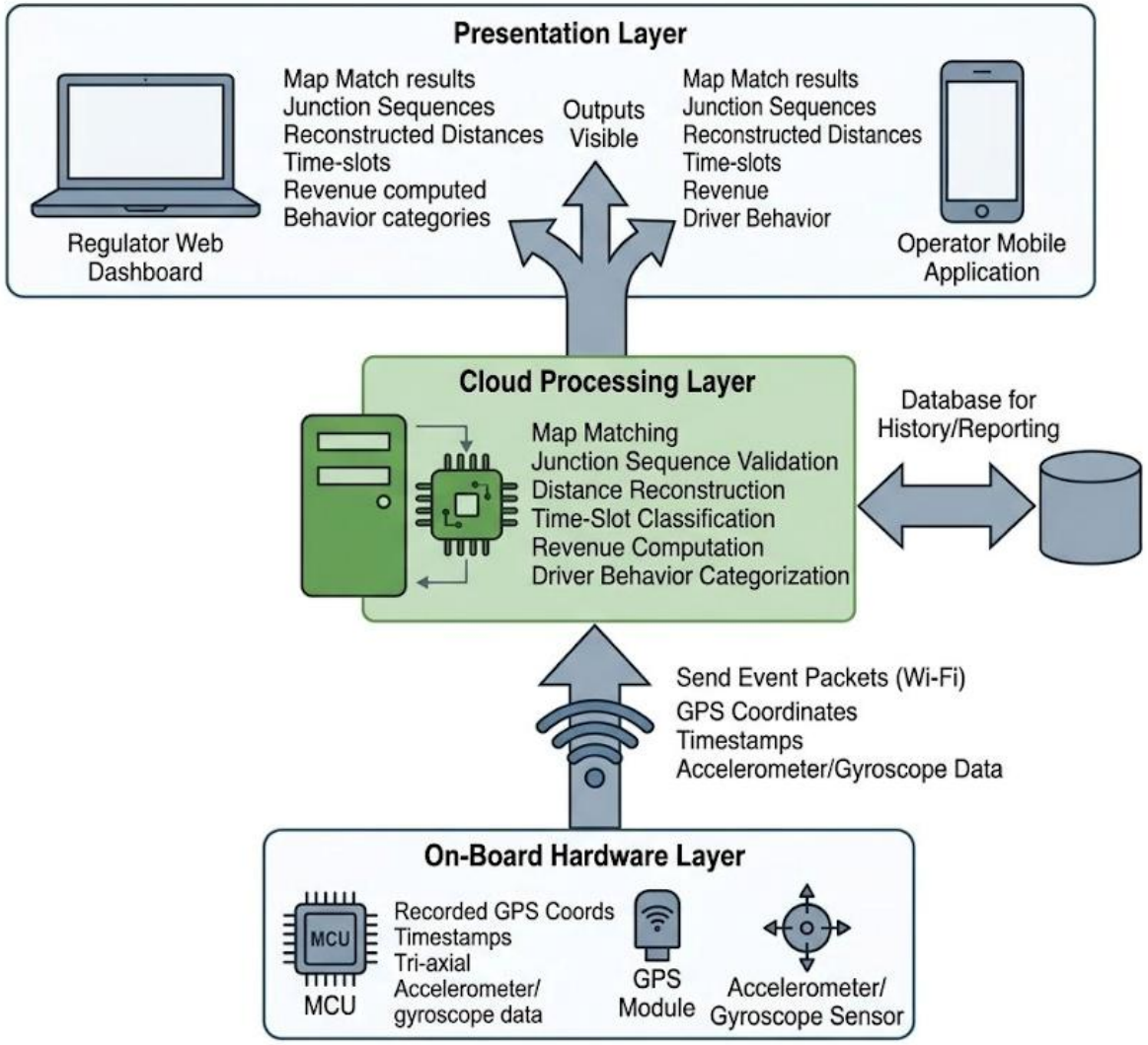
- Fallows a Design Science Research
- Develop an IoT Prototype

Validation

- All participants are informed
- Compares system results with manual
- Vehicles and drivers are identifies using anonymous labels
- Crypted communication is used when transmitting data to ensure security
- DB access is limited to authorized users
- Collects minimum data required for operations

Distance-based Revenue Management Component

System Overview Diagram



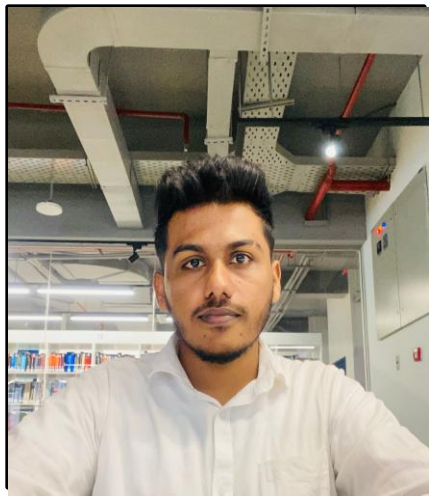
Research Gap and Novelty

Research Gap

- Route permit payments are a fixed amount for every route.
- Existing systems focus on bus tracking only.
- Off-peak operators are able to get paid without reliable service .

Novelty

- An on-board device that can track vehicles' location, speed and aggressiveness .
- Daily distance-based permit fee generator .
- Automatic commission generator for owners operating during off-peak times.



SACHINTHA H D D | IT22264152

BSc (Hons) in Information Technology Specializing in Information Systems Engineering

Policy-Driven Bus Scheduling and Time-Slot Management Engine

Research Problem:- How can a policy-driven time-slot engine automate bus scheduling, and dynamically adjust bus services based on real-time operational events to improve service reliability and policy enforcement?

Specific Objective

To design and implement a Policy & Time-Slot Engine that automatically classifies bus operations into predefined service periods and applies configurable transport policies to support automated scheduling.



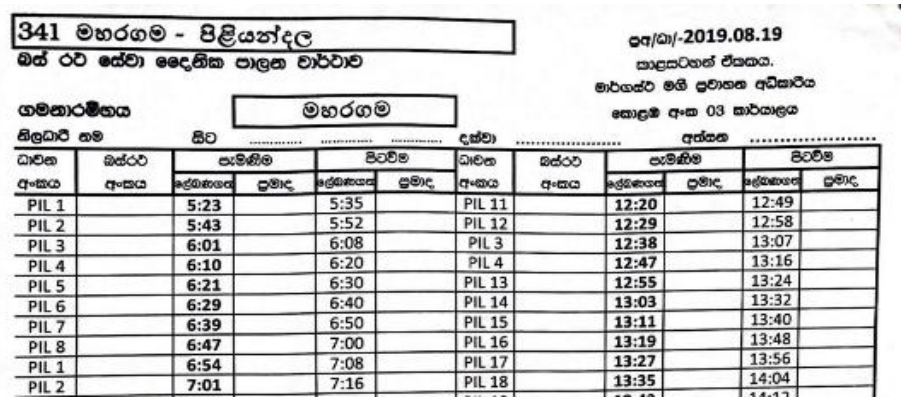
341 පිළියන්දල - මහරගම

ධාවන අංකයන්ට අදාල ධාවන සැලැස්ම 2019.08.19

PLI 13	පිළියන්දල	පැමිණීම	12:17	13:47	15:23	16:59	18:35	####	10
		පිටවීම	12:32	14:08	15:44	17:20	19:20		75
	මහරගම	පැමිණීම	12:55	14:31	16:07	17:43	19:43		4
		පිටවීම	13:24	15:00	16:36	18:12	20:10		8:16
PLI 14	පිළියන්දල	පැමිණීම	12:25	13:55	15:31	17:07	18:43	####	10
		පිටවීම	12:40	14:16	15:52	17:28	19:40		75
	මහරගම	පැමිණීම	13:03	14:39	16:15	17:51	20:03		4
		පිටවීම	13:32	15:08	16:44	18:20	20:30		8:28
PLI 15	පිළියන්දල	පැමිණීම	12:33	14:03	15:39	17:15	18:51	####	10
		පිටවීම	12:48	14:24	16:00	17:36	20:05		75
	මහරගම	පැමිණීම	13:11	14:47	16:23	17:59	20:28		4
		පිටවීම	13:40	15:16	16:52	18:28	20:50		8:40
PLI 16	පිළියන්දල	පැමිණීම	12:41	14:11	15:47	17:23	18:59		8
		පිටවීම	12:56	14:32	16:08	17:44			60
	මහරගම	පැමිණීම	13:19	14:55	16:31	18:07			3
		පිටවීම	13:48	15:24	17:00	18:36			6:18

Sub Objectives

- Develop a time-slot classification mechanism.
- Design a policy rule management module.
- Implement an automated schedule adjustment mechanism.



341 මහරගම - පිළියන්දල

ඔස් රථ සේවා දෛනික පාලන වාර්තාව

ප්‍ර/ධා/-2019.08.19

ගමනාරම්භය මහරගම

ධාවන අංකය	බස්රථ අංකය	පැමිණීම නියමිත වේලාව	පැමිණීම ප්‍රමාද	පිටවීම නියමිත වේලාව	පිටවීම ප්‍රමාද	ධාවන අංකය	බස්රථ අංකය	පැමිණීම නියමිත වේලාව	පැමිණීම ප්‍රමාද	පිටවීම නියමිත වේලාව	පිටවීම ප්‍රමාද
PIL 1		5:23		5:35		PIL 11		12:20		12:49	
PIL 2		5:43		5:52		PIL 12		12:29		12:58	
PIL 3		6:01		6:08		PIL 3		12:38		13:07	
PIL 4		6:10		6:20		PIL 4		12:47		13:16	
PIL 5		6:21		6:30		PIL 13		12:55		13:24	
PIL 6		6:29		6:40		PIL 14		13:03		13:32	
PIL 7		6:39		6:50		PIL 15		13:11		13:40	
PIL 8		6:47		7:00		PIL 16		13:19		13:48	
PIL 1		6:54		7:08		PIL 17		13:27		13:56	
PIL 2		7:01		7:16		PIL 18		13:35		14:04	

Policy-Driven Bus Scheduling and Time-Slot Management Engine

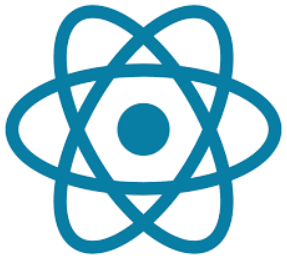
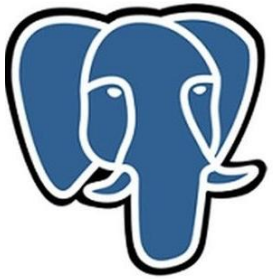
Methodology

Design Science Research for intelligent transport scheduling: a rule-based engine processes trip timestamps, applies policy rules, performs time-slot classification, and dynamically adjusts bus schedules during operational disruptions.

Technologies

- Azure IoT Hub
- React JS
- Spring Boot
- PostgreSQL



Policy-Driven Bus Scheduling and Time-Slot Management Engine

How to Validate :

- **Operational Data Validation:** Verify route No, trip start time, and operational events using real bus trip datasets.
- **Time-Slot Classification Validation:** Compare system-assigned time slots (Peak, Normal, Night) with predefined policy definitions.
- **Schedule Adjustment Validation:** Simulate delays or breakdowns and verify automatic reassignment of the next available bus.
- **Performance Validation:** Measure system response time and scheduling decision speed under multiple simulated operational scenarios.

Expected Accuracy

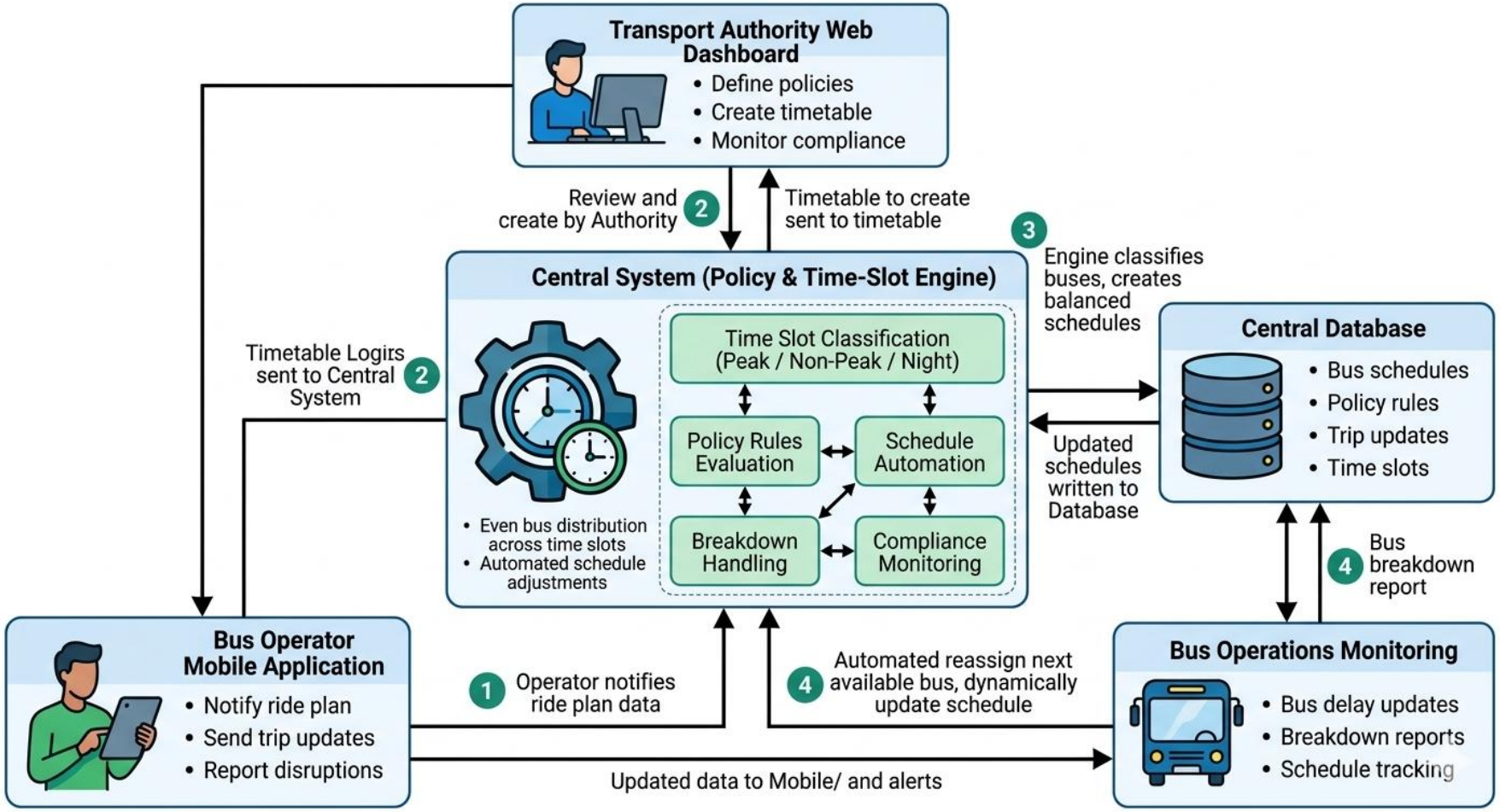
- Time-Slot classification Accuracy: 95% or higher
- Automated Schedule Adjustment Reliability: 90-95%
- Scheduling Decision Response Time: < 2 Seconds
- Overall System Accuracy: 90-95%

Data Availability

- Operational trip data, including route Number, trip timestamps, service events, and disruption records, will be collected from transport monitoring systems and simulated operational datasets.
- Reference data such as route schedules and operational policies will be obtained from the **Road Passenger Transport Authority** and related transport management sources.

Policy-Driven Bus Scheduling and Time-Slot Management Engine

System Overview Diagram



Research Gap and Novelty

Research Gap

- Bus schedules in Sri Lanka **are prepared manually.**
- Existing systems focus on **bus tracking, not automated.**
- **No policy-based system** to classify services into peak, normal, and nighttime slots.
- Difficult **to adjust schedules** when delays or bus breakdowns occur.

Novelty

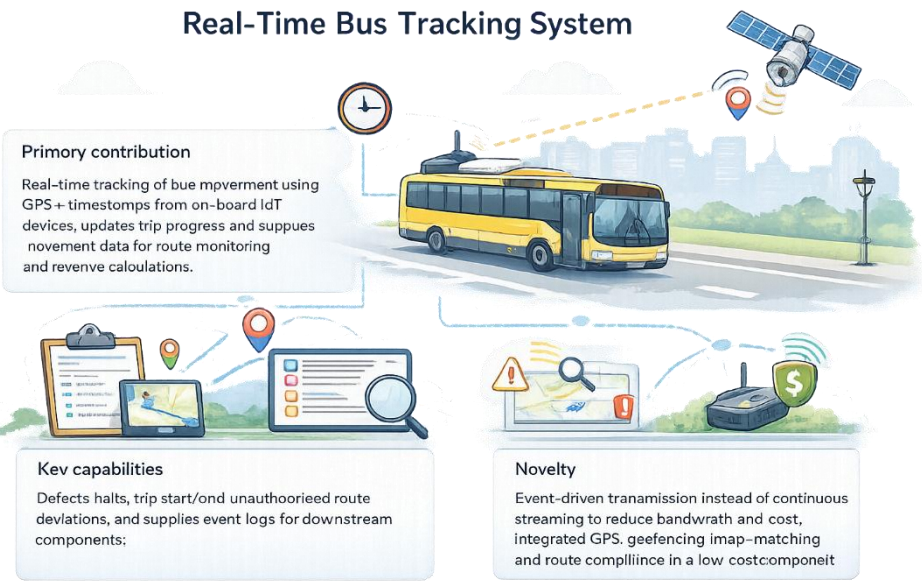
- **Policy-driven automated** bus scheduling engine.
- **Automatic time-slot classification.**
- **Dynamic schedule adjustment** during delays or breakdowns.
- Supports **transport authority policy enforcement and monitoring.**

Event-Driven Bus Monitoring Engine Engine



IT22121042 MIFLAL AHAMED M.T

BSc (Hons) in Information Technology Specializing in Information Systems Engineering



Primary contribution

Real-time tracking of bus movement using using GPS + timestamps from on-board IoT on-board IoT devices; updates trip progress progress and supplies movement data for for route monitoring and revenue calculations.

Key capabilities

Detects halts, trip start/end, unauthorized unauthorized route deviations, and supplies supplies event logs for downstream components.

Novelty

Event-driven transmission (instead of continuous streaming) to reduce bandwidth and cost; and cost; integrated GPS, geofencing, map-matching, and route compliance in a low-cost low-cost component.

Problem, Objective, and Sub-objectives



Problem definition

- Unauthorized route deviations go undetected by many fleets.
- Continuous GPS streaming increases communication cost and power usage.
- Existing systems focus on passenger information rather than route compliance enforcement.

Research objective

Design an event-driven processing engine that updates bus location and trip progress using intermittent GPS + timestamp events from on-board IoT devices.

01

Capture

Collect coordinates and timestamps via GPS modules on ESP32-based devices. devices.

02

Detect events

Identify trip start/end, halts, halt-to-halt movement, and route deviations.

03

Map & validate

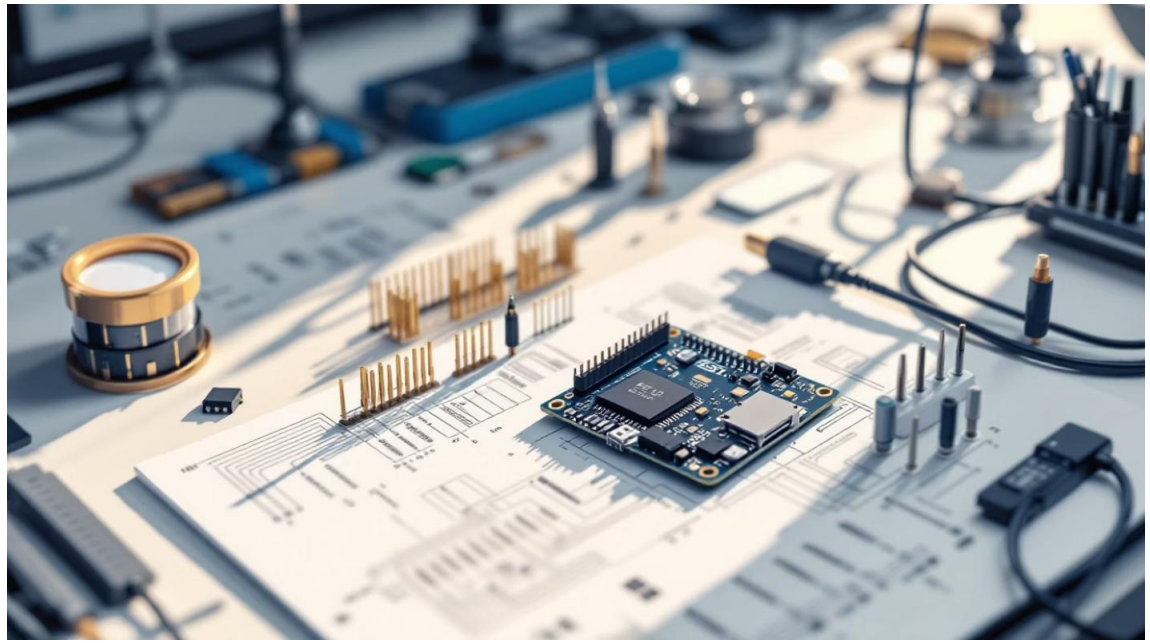
Apply map-matching against official route polylines and geofence checks at at junctions/halts.

04

Reporting

Flag tamper/offline conditions and produce event feeds for policy, revenue, and revenue, and mobile components; validate via field tests.

Methodology, Technologies & Validation



Methodology

Design Science Research in intelligent transport; algorithmic pipeline combining geofencing, Haversine Haversine distance checks, and map-matching for route for route compliance detection.

Technologies

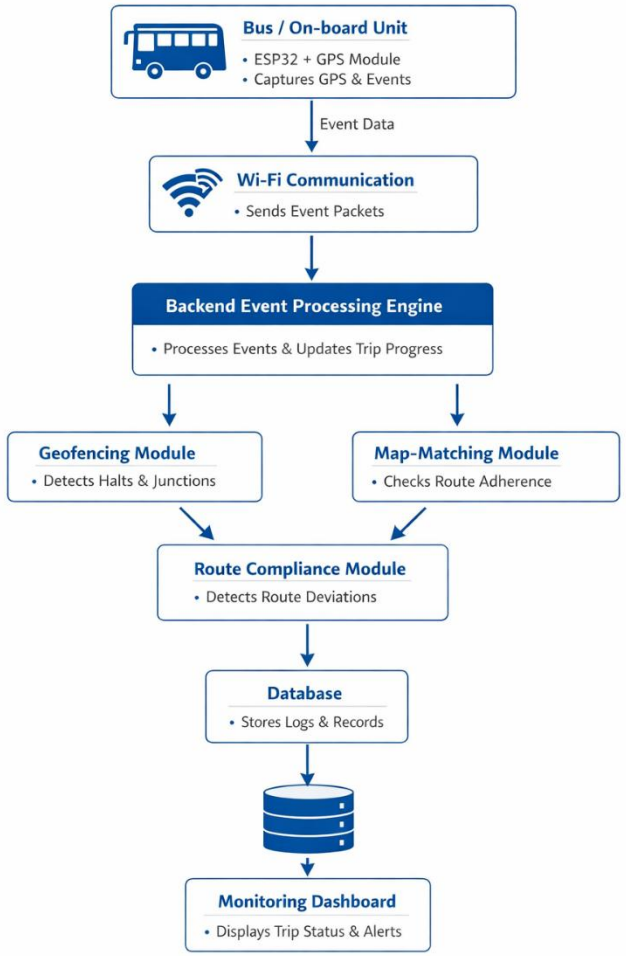
- ESP32 microcontroller (device logic, event generation)
- NEO-6M GPS module (coordinates + UTC timestamps)
- Wi-Fi event transmission (batched/event-driven)
- Backend: route polyline store, map-matching engine, event processor

Validation metrics

- GPS positional accuracy (median error)
- Junction detection precision & recall
- Route deviation detection accuracy (true/false positives)
- Device reliability and end-to-end latency

Data collected are GPS coordinates, timestamps, and route metadata. Access must be restricted; field testing requires informed consent from drivers and drivers and operators.

High-Level Architecture diagram



System Overview — Inputs, Process, Outputs



Inputs

GPS coordinates, timestamps, Bus ID, route polylines, junction/halts list



Processes

Event detection, geofence checks, map-matching, route route compliance and tamper/offline detection



Dependencies

On-board IoT hardware; shared backend route DB; DB; policy, revenue, and mobile app modules

Outputs

- Real-time trip progress updates (event stream)
- Halt detection and halt-to-halt movement events
- Route deviation and unauthorized route change alerts
- Event logs for revenue calculation and policy enforcement

Design supports modular integration: event API feeds policy engine and revenue reconciler; events are c events are compact to minimize bandwidth and storage.

IoT-Integrated Driver Monitoring and Operational Reporting Mobile Application Application



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BSc (Hons) in Information Technology Specializing in Information Systems Engineering

Primary contribution

Development of an IoT-integrated mobile application that combines real-time driver behavior monitoring with operational and financial visibility. The application uses on-board sensor data such as GPS, speed, acceleration, gyroscope, and timestamps to detect unsafe driving events and generate instant alerts, while also providing trip summaries, trip status updates, bus availability reporting, breakdown notifications, driver behavior reports, bus schedules, earnings details, and route-related payment information for drivers and bus owners..

Key capabilities

Monitors driver behavior in real time using GPS, speed, acceleration, gyroscope, and timestamp data; detects inefficient driving events, generates immediate through the mobile application, and enables reporting including bus availability, trip status, and reporting..



Novelty

Real-time driver behavior monitoring using IoT sensor data to detect unsafe driving events such as overspeeding, aggressive braking, aggressive turning, and abnormal slow movement.

Provides instant mobile alerts while improving owner visibility, reducing accident risk, and enhancing passenger safety and comfort

Problem, Objective, and Sub-objectives



Problem definition

- No effective real-time driver behavior monitoring.
- Unsafe driving events are hard to detect live
- Limited owner visibility into trips, schedules, earnings, and payments.
- No integrated mobile platform for monitoring and reporting.

Research objective

To develop an IoT-integrated mobile application for real-time driver monitoring, operational reporting, and owner-side visibility using sensor data, backend analytics, and intelligent mobile alerts..

01

Capture

Collect GPS, speed, acceleration, gyroscope, and timestamp

02

Detect

Identify unsafe driving events in real time

03

Alert

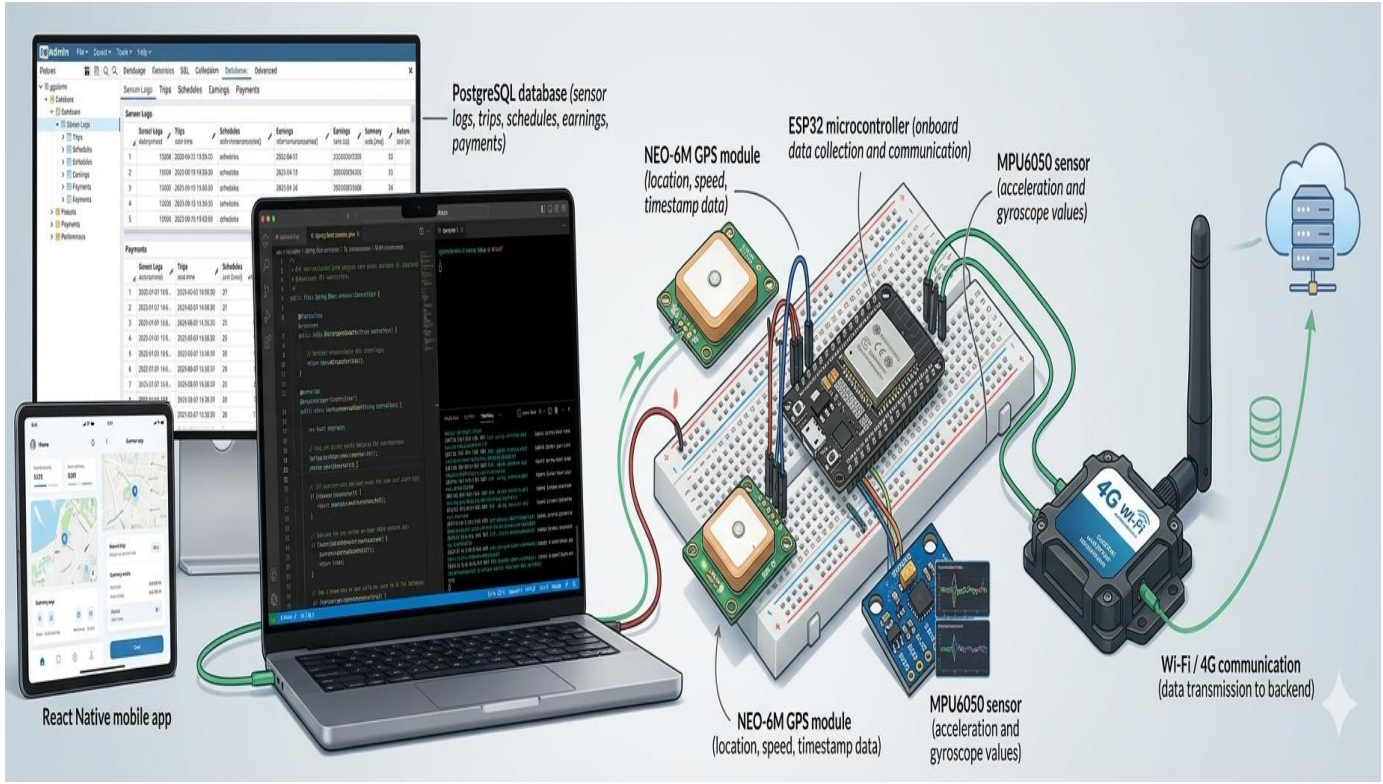
Send instant alerts and provide monitoring information

04

View

Show trip summaries, schedules, earnings, and payment details

Methodology, Technologies & Validation



Methodology

Design Science Research approach for developing an IoT-integrated mobile application; real-time data collection using onboard sensors; sensor-data analysis for driver behavior detection; backend processing for alert generation, operational reporting, trip summaries, and owner-side visibility.

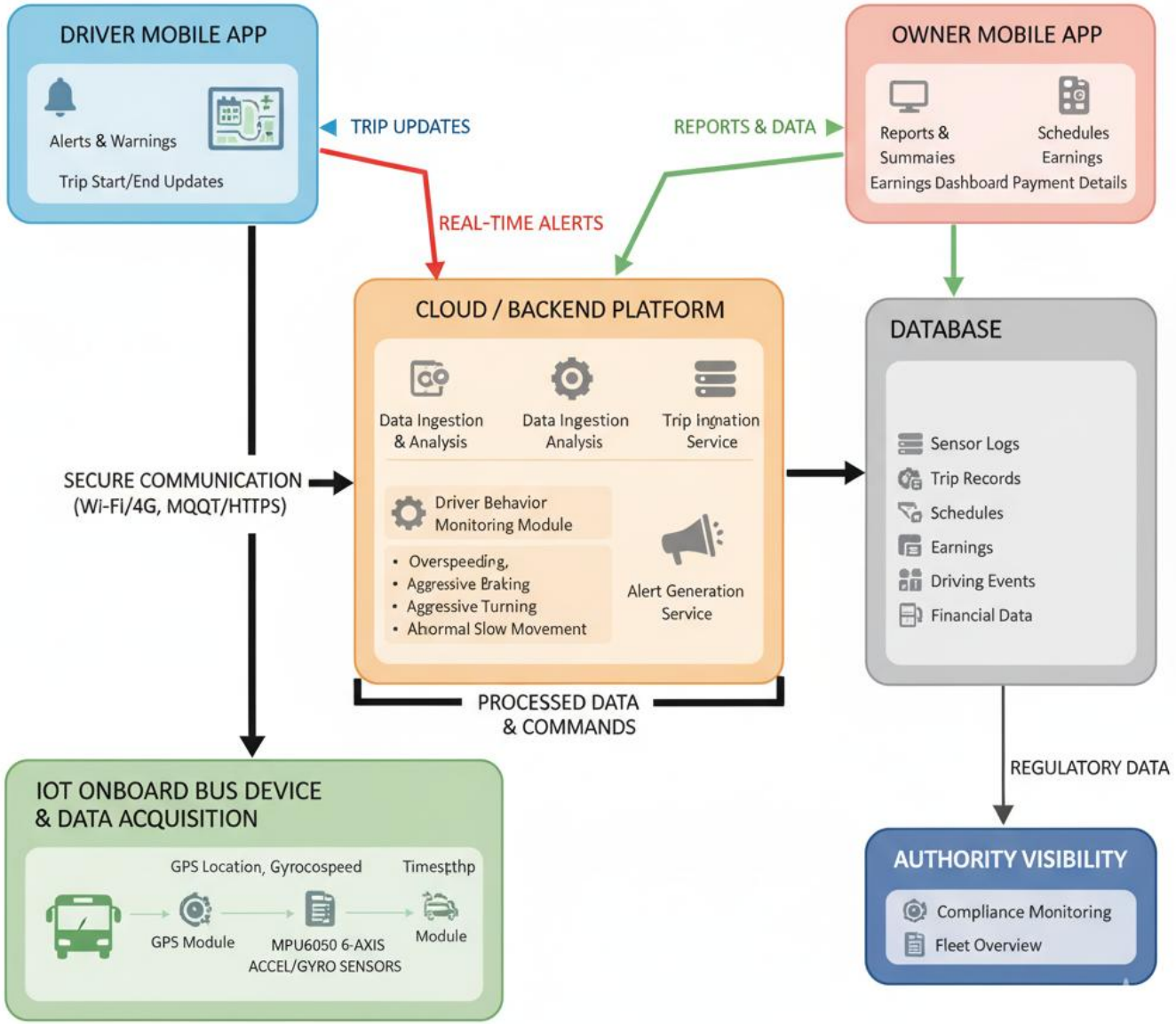
Technologies

- **ESP32 microcontroller** (onboard data collection and communication)
- **NEO-6M GPS module** (location, speed, timestamp data)
- **MPU6050 sensor** (acceleration and gyroscope values)
- **Wi-Fi / 4G communication** (data transmission to backend)
- **Spring Boot backend** (data processing, APIs, alert logic, reports)
- **PostgreSQL database** (sensor logs, trips, schedules, earnings, payments)
- **React Native mobile app** (driver and owner mobile interface)

Validation

- Compare detected events with actual verified events
- Measure overspeed, braking, and turning detection accuracy.
- Aggressive braking / turning detection accuracy
- Real-time alert response time/mobile app
- Mobile app reporting accuracy
- Data transmission reliability and end-to-end latency
- Accuracy of trip summaries, schedules, earnings, and payment details

System Overview Diagram



IoT-Integated Driver Monitoring



- Focus is mostly on bus tracking, not driver behavior monitoring
- No integrated real-time unsafe driving detection
- No instant driver alert mechanism
- No combined driver/owner mobile platform
- No trip summary and behavior reporting integration
- Gap: lack of one integrated IoT mobile solution for monitoring, alerts, reporting, and visibility

Commercialization



Market & customers

Target: municipal transit authorities, private bus operators, paratransit services seeking compliance and compliance and cost reduction.

Value proposition

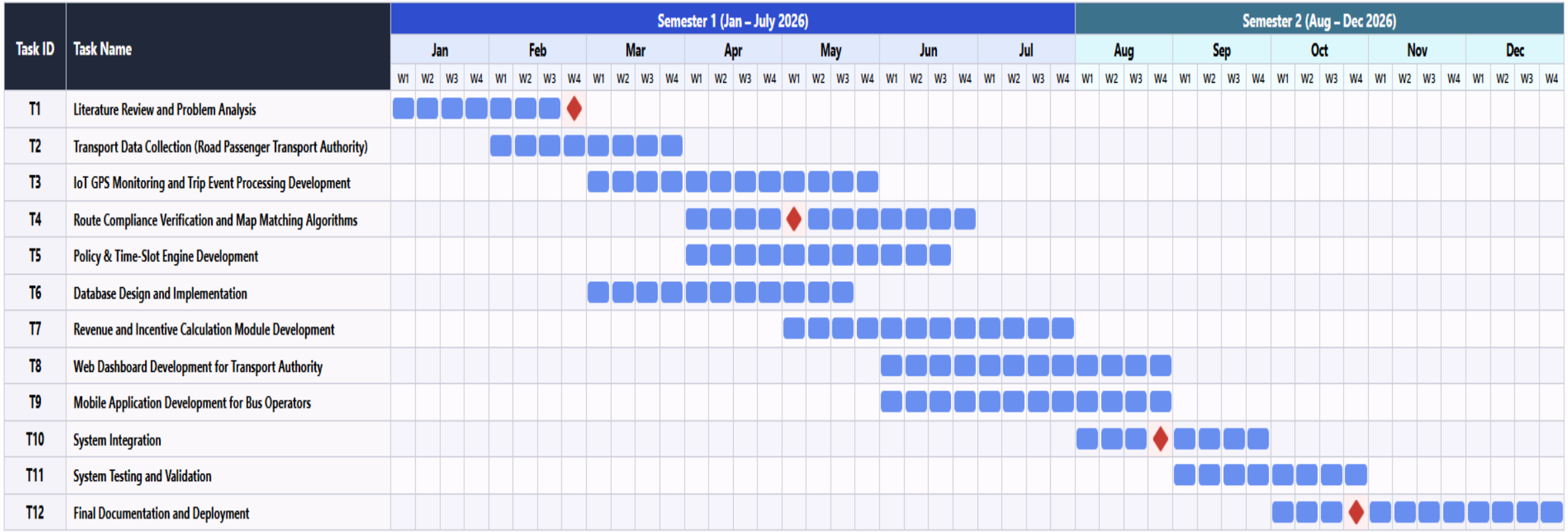
- Reduced communication costs via event-driven driven reporting
- Automated route compliance and tamper detection
- Integration-ready event feed for revenue and policy systems

Pricing & IP

Low-cost device + subscription for backend processing; processing; potential for software IP (map-matching (map-matching optimizations) and deployment services.

Investment primarily in device provisioning, backend scaling, and field validation. Early adopters can pilot with a small fleet to validate ROI.

Research Project Gantt Chart — 2026

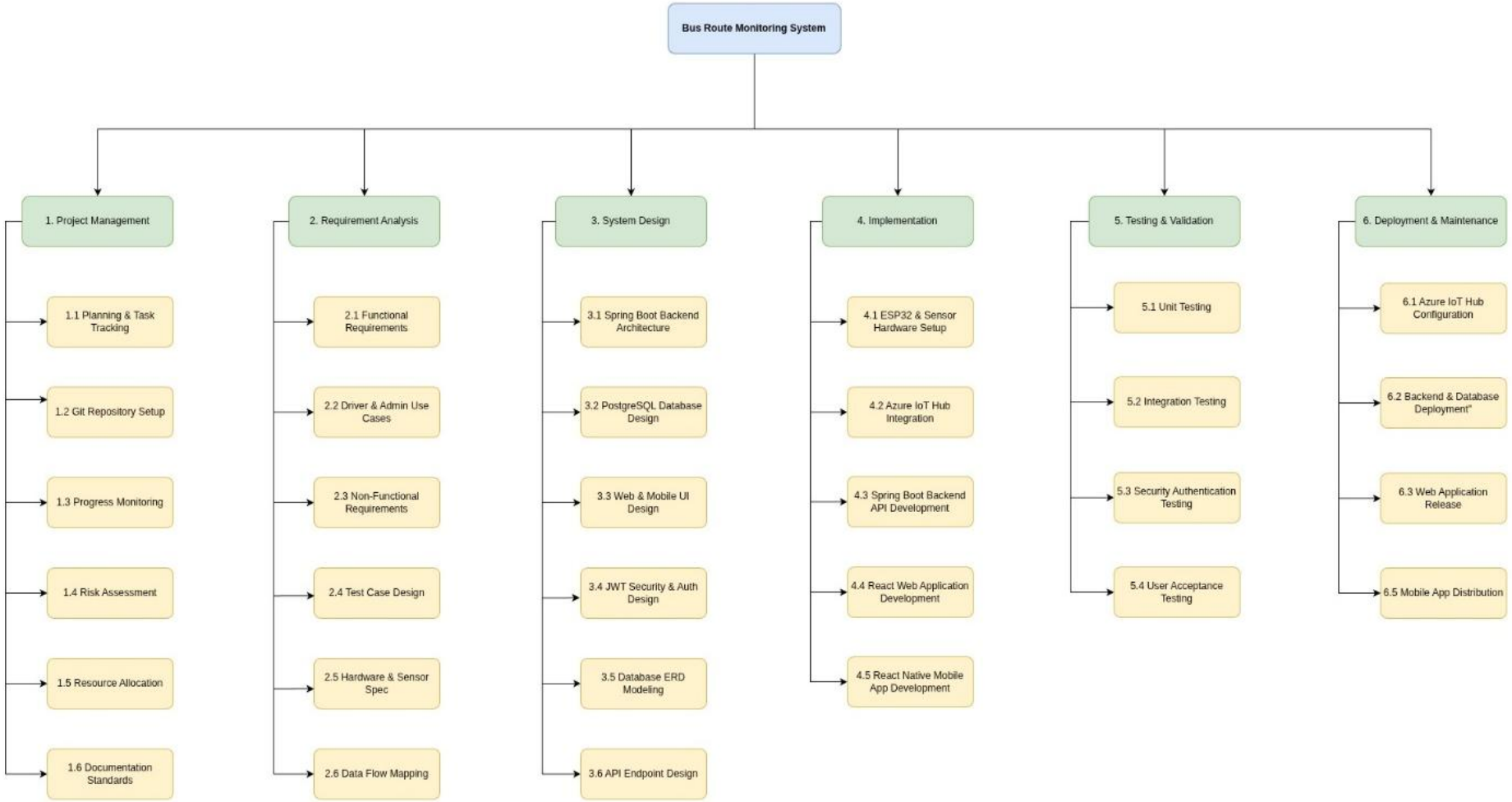


Key Milestones

- Month 2 – Dataset Preparation Completed
- Month 4 – Core System Components Developed
- Month 8 – System Integration & Testing Completed
- Month 10 – Final Documentation & Deployment

Legend: Task in Progress, Milestone

Work Breakdown Chart



Thank You

